

Product: **CycurLIB**

Version: **3.5**

Release Date: **2018-02-28**

API Compatibility

The API of CycurLIB 3.5 was changed for the following algorithms:

- ECIES
 - The length of the plaintext is now variable at runtime. CycurLIB 3.4 fixed the length at compile-time.
- CMAC
 - The CMAC implementations were extended by an additional Start() function such that the MAC computation can be re-started without executing the key schedule again (done inside Init()). Calling Start() is not mandatory, i.e. the change is completely backward compatible to earlier versions.

The following insecure legacy algorithms were removed:

- MD2
- MD5
- DES
- ARC4

New Features

The following algorithms were added with CycurLIB 3.5:

- AES-EAX mode
- Key wrapping for AES
- A key derivation function that can be used for e.g. deriving keys from an ECDH shared secret
- Miyaguchi-Preneel compression and key derivation according to the Secure Hardware Extension (SHE) standard

Improvements

The performance of the ECC core implementation was improved in this release. In particular, signature generation is faster with the new fixed base option (see `EscEcc_USE_FIXED_BASE` in `ecc_cfg.h`).

Legacy Algorithms

The following algorithms are only included in the CycurLIB to support legacy applications. They are **INSECURE** and should not be used for new products:

- SHA-1

Please refer to the manual for alternative algorithms.

Known issues and limitations

Known limitations

- The development of CycurLIB follows a process that is not compliant to ASPICE or similar standards.

Known bugs

<none>